MUDITHA PRIYASAD

Mechanical & Automation Engineer | Machine Design | Factory Automation | Troubleshooting | Prototyping

+94 71 7599 667 | muditha00@icloud.com | Thambuttegama, Sri Lanka | linkedin.com/in/muditha00
IESL Membership: AM-32795 | Portfolio: Available on request

PROFESSIONAL IDENTITY

Hands-on Mechanical and Automation Engineer with 5+ years of study, internships and industry experience designing, building, commissioning and troubleshooting practical factory machines, jigs, fixtures, feeder mechanisms and prototype systems. Strong ability to connect mechanical design, pneumatics, PLC-based automation, sensors, 3D printing, CNC/laser systems and digital tools to solve real production problems. Known for adapting new technology and converting ideas into workable engineering solutions.

ENGINEERING SNAPSHOT

2+	**15+**	**4**	**2**
Years of Experience	Practical Projects	Industrial Environments	SOLIDWORKS Certifications

WHAT I BRING

Design practical machines: Jigs, fixtures, feeders, frames, mechanisms and production-support tools.	**Solve factory problems:** Troubleshoot mechanical, pneumatic, electrical, sensor and control-related faults.
Build and test prototypes: Use SOLIDWORKS, 3D printing, CNC laser systems and workshop fabrication.	**Support production reliability:** Commission machines, identify root causes and reduce repeated issues.
Adapt new technology: Raspberry Pi, Arduino, AI tools, sensors and ERP/workflow concepts.	**Work across disciplines:** Mechanical, electrical, pneumatic, control, fabrication and production teams.

CORE SKILLS

Mechanical Design:	SOLIDWORKS, 3D/2D CAD, fabrication drawings, mechanism design, jigs, fixtures, machine frames, DfMA/DfAM, material and process awareness
Automation & Controls:	PLC programming (Delta, Xinje, Mitsubishi, Siemens), sensors, actuators, pneumatics, servo mechanisms, electrical wiring, control panels, SCADA fundamentals, commissioning support
Troubleshooting & Maintenance:	Industrial machine troubleshooting, root cause analysis, mechanical evaluations, preventive maintenance support, TPM, 5S, Kaizen, production reliability
Prototyping & Digital Tools:	3D printing, CNC laser engravers, G-code workflows, linear feeder/orientation mechanisms, Raspberry Pi, Arduino, Python, C++, Microsoft Office, AI-assisted ERP/workflow concepts

PROFESSIONAL EXPERIENCE

Mechanical Design & Automation Engineer | **MARQO Industries Pvt Ltd** | Jan 2024 – Present

- Design, build and commission factory automation machines by combining mechanical design, PLC control, sensors, pneumatics, wiring, fabrication support and system integration.
- Work on LED bulb assembly, concrete/fan-base production support, eyelet punching, plug-top packing, spot welding, screw feeding and linear feeder/orientation systems.
- Prepare SOLIDWORKS concepts, fabrication guidance and prototype parts for jigs, fixtures, feeders, machine frames and production-support tools.
- Troubleshoot mechanical, pneumatic, electrical and control faults; identify root causes and apply practical corrective actions to reduce repeated failures.
- Use 3D printing and rapid prototyping to verify fit, assembly access, mechanism function and manufacturability before fabrication or modification.
- Apply AI tools (ChatGPT, Claude) to develop ERP/workflow concepts covering inventory, approval flows, KPI dashboards, task visibility and cloud-ready data planning.
- Support planning and setup direction for a functional automation workshop and QC lab.

Founder / Designer | **MTone Designers** | 2021 – Present

- Founded and operate an engineering design venture focused on 3D printing, CNC laser engraving, wood-burning artwork and custom machine concepts; serving individual and small-business clients.
- Develop custom 3D printer and CNC laser engraver concepts covering mechanical frames, motion systems, electronics integration, G-code operation and machine testing.

Engineering Intern | MAS **Active Shadowline** | 2024 (9 months)

- Supported 5S, Kaizen and TPM activities, service-room planning, production-support documentation, CAD work, department dashboards and energy/water tracking.

Automation Engineering Intern | **Effective Engineering & Automation Pvt Ltd** | 2023 (6 months)

- Gained hands-on experience in industrial automation, maintenance troubleshooting, electrical wiring, machine design, PLC/SCADA programming and documentation.
- Worked with equipment and projects including ball mill, tensile tester, combustion chamber, 3D printer, line follower robot, screen printer, concrete mixers and glove-making robot.

Automation Engineering Intern | **MAS Kreeda Vaanavil** | 2019 (3 months)

- Gained practical exposure to pneumatic circuit design, CAD design, PLC ladder programming, Arduino programming, power tools and lathe-machine operation.
- Supported boiler monitoring, Ottuthal bonding-tape machine development and fabric-picking mechanism research.

SELECTED PROJECT EVIDENCE

Factory Automation Machines (MARQO):	LED bulb assembly, concrete/fan-base production support, eyelet punching, screw feeding, plug-top packing, spot welding and production troubleshooting systems.
Linear Feeding & Orientation Mechanisms:	Component feeding, orientation, transfer, sensing and integration with repetitive production-machine workflows.
CNC Laser Engraver & Custom 3D Printer:	Machine-development work covering mechanical frames, motion systems, electronics integration, G-code workflows, testing, operation and Raspberry Pi based control concepts.
Robotics & Mechatronics:	Bio-inspired hopping robot, line follower robot with object pickup/lifting/colour detection, combat robots and 8×8×8 LED cube.
Practical Semi-Automated Machines:	Pneumatic wire stripper, CNC PCB drilling machine, gripper fixture, fabric-picking mechanism, Ottuthal machine, oil dripping tray and screen printer machine concepts.

CAREER VALUE SUMMARY

Strongest in practical engineering environments where machines need to be designed, modified, tested, repaired or improved. Value comes from connecting mechanical design, automation, troubleshooting, production needs and low-cost prototyping to solve real factory problems.

EDUCATION & CERTIFICATIONS

BSc (Hons) in Mechanical Engineering Faculty of Engineering, University of Jaffna | 2018 – 2023 **GCE A/L – Physical Science** Anuradhapura Central College | 2013 – 2016	**Certifications & Training** CSWA – CAD Design CSWA-AM – Additive Manufacturing PLC Programming Course | Epic Engineering | 2024 PLC Training | University of Moratuwa Maintenance of Industrial Plant & Machinery | NERDC Industrial Motor Operation & Control System | NERDC Advanced Injection Molding | IDB

ACHIEVEMENTS & ADDITIONAL INFORMATION

Bronze Award for University Stall with the Best Display of Engineering Projects, Techno Sri Lanka | 2023. Participant, AIR TATTOO Exhibition | 2024; Organizer, UOJ BATTLEGROUNDS Exhibition Competition | 2020; Participant, SLT Robot Battles | 2019.

Languages: Sinhala and English. **References:** Available on request.